

One advantage that makes up for the lower resolutions in the latest ATi models is the enhanced anti-aliasing output that CrossFire offers. Basically, the two cards render frames with different anti-aliasing levels, and then the compositing chip puts them together using a new technique called Adaptive AA (Anti-Aliasing).

Adaptive AA is ATi's new method of improving graphic quality. Basically, it uses previous methods of multiple AA sampling, but intelligently.

What this means is that so far, chips were taking multiple samples of almost everything with a texture, and then sampling it multiple times to get the best quality textures. This sampling is done by shifting the centre of a pixel around and then rendering it. When done multiple times, the outputs are blended together to give you a higher resolution texture.

This, however, is not very efficient, as not all textures in the display need to be high resolution to output great graphics. This is where Adaptive AA comes in. It intelligently decides which textures will improve the visual output and which won't. The textures that need multiple rendering passes (to make the output look better) are given priority, thus increasing the visual detail.

Single Card Multi-GPU

Both nVidia and ATi offer dual GPUs on a single card. These multi-GPU cards offer better performance, albeit only theoretically. Our various graphics card tests have shown us that dual-PCIe cards perform better than a single dual-GPU card—two 6800 GTXs in SLI mode outperformed a 6800 Ultra dual-GPU card.



Card makers have started putting two GPUs on cards to get the best performance possible. It gets better when you use two of these dual-GPU cards in SLI or CrossFire mode



Extreme Hardware

For gaming freaks, there's also the option of nVidia's Quadro and ATi's FireGL range of workstation cards, which, in all honesty, we have ignored through all our articles thus far. Why? For starters, they are way too expensive!

These cards, (Quadro, FireGL, and even Creative's Oxygen line) are used for really high-end video rendering, in systems that costs lakhs of rupees. They support ridiculous levels of full screen anti-aliasing (32x) and offer multiple screen outputs. nVidia's cards can also be run in SLI mode, thus doubling their already superlative performance.

Overall, these are not mass consumer gaming cards—not that the Rs 30,000+ nVidia and ATi cards can be bought by everyone anyway, but they're cheaper than workstation solutions.

For gaming addicts with money to burn, there are many more performance solutions out there: the Gigabyte GA-8N SLI Quad Royal motherboard, for example. This board offers four, yes, *four*, PCIe graphics card slots, though you should know that nVidia's drivers do not currently support four GPUs.

You could also overkill the idea of GPUs and plug in four 7800 GTs, which makes a total of *eight* GPUs running on your system. It's kind of pointless, since all four PCIe graphics ports running together only offer 8x PCIe speeds per port.

Thus your 7800 GTs would be running considerably under power, but with a glass cabinet, a little in-cabinet lighting and one monstrous power supply, you could be the talk of the gamer clan across the world!



The Quad Royal is the perfect example of extreme hardware—four cards in SLI mode!

There's a limitation in the motherboard's PCIe chipset architecture that reads two PCIe x16 slots as two PCIe x8 slots—due to the fact that the chipsets only support about 20 PCIe pipelines. Thus, even two graphics cards installed on two PCIe x16 slots, running in CrossFire or SLI mode, are used at x8 PCIe speeds each.

When compared to a single dual-GPU card, which should be detected as a full fledged x16 PCIe card, it's obvious that the single dual-GPU card should outperform the dual-card setup. In the real world, this just isn't true, and dual-card setups are faster—as of now! This is still so because motherboard chipset makers are still trying to catch up with the graphics card market in terms of speed and technology.

Currently, nVidia's flagship GPU is the 7800 GTX, while ATi launched the X1800 very recently. Both these chips will be used to create multi-GPU cards, which offer the ultimate in graphics technologies. We haven't had a chance to compare them in a direct shootout yet here at Digit, but we will, soon.

